FORENSICS IN DIGITAL SECURITY (FDS)

Data Storage In Digital Format I

OBJECTIVES

- Describe digital evidence
- Describe forensic process
- Prepare for forensic investigation
- Describe search and seizure during investigation
- Analyze evidence of investigation
- Report, testify and close investigation

DIGITAL EVIDENCE

"Digital object that contains *reliable information* that *supports or refutes a hypothesis*"

In legal terms, "any *probative information* stored or transmitted *in digital form* that a party to a court case *may use at trial*."

- Brooks, C. L.
CHFI Computer Hacking Forensic Investigator Certification All-in-One Exam Guide

Digital evidence is information stored or transmitted in binary form that may be relied on in court.

- National Institute of Justice(NIJ)
Research, development and evaluation agency of the U.S. Department of Justice

DIGITAL EVIDENCE

Authenticity
- Where does the evidence come from?

Reliability
- Is the evidence reliable and free of flaws?

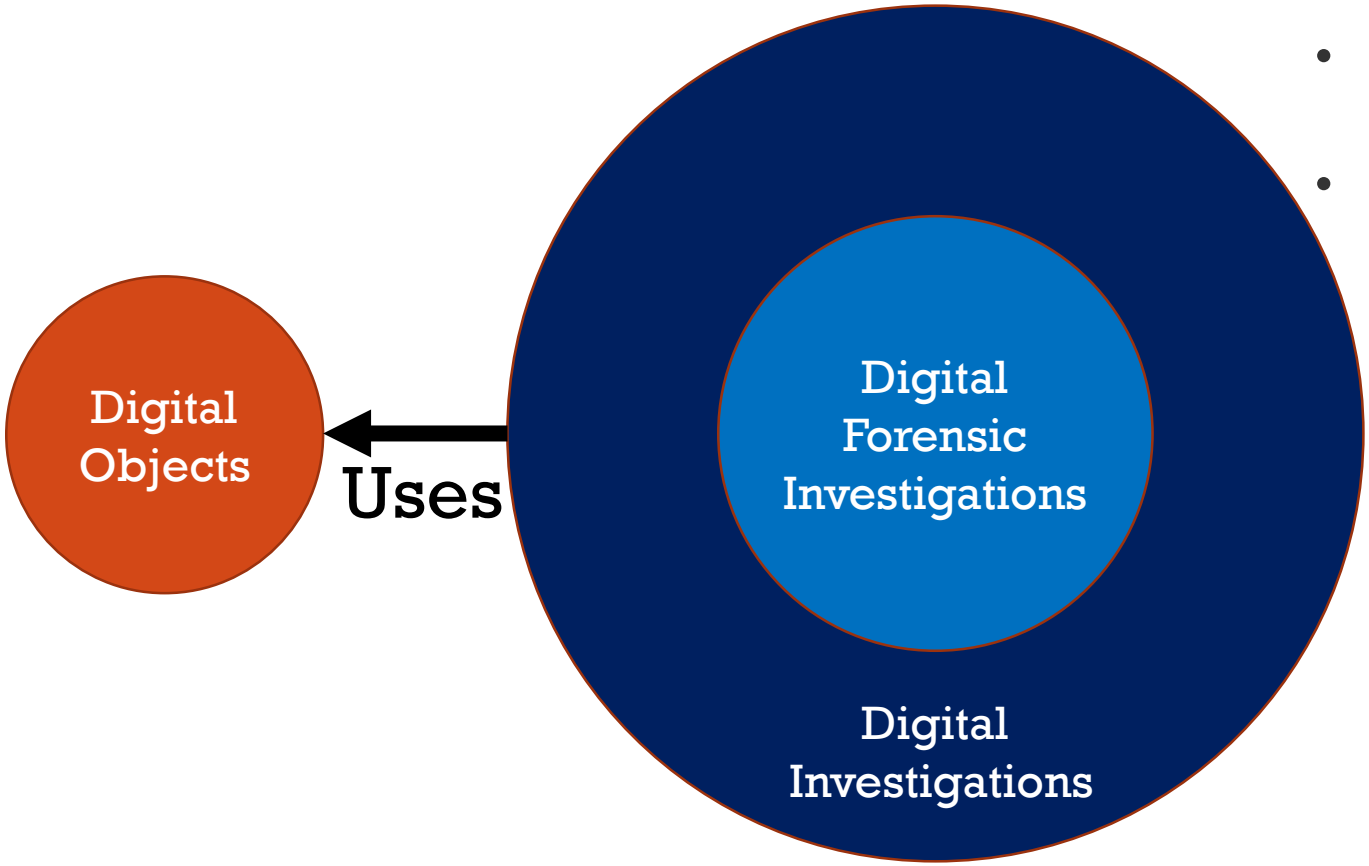
Admissible
- Does evidence conform to legal rules for courts?

Complete
- Does evidence tells the whole story?

Believable
- Is evidence readily available and understandable by court?

Guidelines for Evidence Collection and Archiving
https://tools.ietf.org/html/rfc3227

DIGITAL INVESTIGATIONS



Digital forensic investigations

- A subset of digital investigations
- More constraints by the forensic process
- Digital evidence admissible in court

Digital objects, digital investigations, and digital forensic investigations

(Adapted from Carrier, B. File System Forensic Analysis (NJ: Pearson Education, 2005), p 4.)

ANTI-DIGITAL FORENSICS

"A set of *methods and tools* that are used to *destroy or obfuscate digital objects* such that they *cannot be used as digital evidence*"

- Brooks, C. L.
CHFI Computer Hacking Forensic Investigator Certification All-in-One Exam Guide

"*Methods undertaken* in order to *thwart the digital investigation* process conducted by legitimate forensic investigators"

- Kevin Conlan, Ibrahim Baggili, Frank Breitinger,
Anti-forensics: Furthering digital forensic science through a new extended, granular taxonomy

Kevin Conlan, Ibrahim Baggili, Frank Breitinger,
Anti-forensics: Furthering digital forensic science through a new extended, granular taxonomy,
Digital Investigation, Volume 18, Supplement, 2016, Pages S66-S75, ISSN 1742-2876,

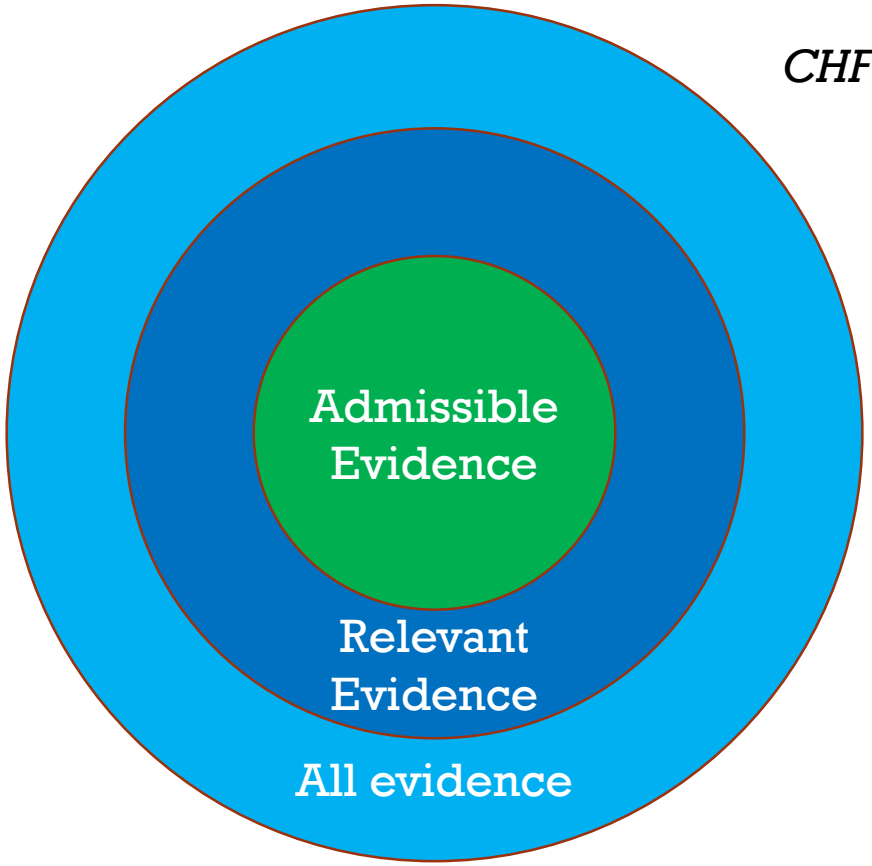
ANTI-DIGITAL FORENSICS

- Encryption
- Steganography
- Packing
- Compression
- Modification of File Metadata
- File Erasure
- Hidden File Systems
- More

FEDERAL RULES OF EVIDENCE (FRE)

"Collection of laws that determine what can or cannot be admitted into evidence in a federal courtroom"

- Brooks, C. L.
CHFI Computer Hacking Forensic Investigator Certification All-in-One Exam Guide



Evidence, relevant evidence, and admissible evidence

(Adapted from CHFI Computer Hacking Forensic Investigator Certification All-in-One Exam Guide Figure 2.5)

INTERNATIONAL PRINCIPLES OF COMPUTER EVIDENCE

Guidelines of Best Practice in the Forensic Examination of Digital Technology

http://enfsi.eu/wp-content/uploads/2016/09/1._forensic_examination_of_digital_technology_0.pdf

- International Organization on Computer Evidence

Publications on audio forensics, cell phone forensics, and general digital forensics
https://www.swgde.org/documents/Current%20Documents

- Scientific Working Group on Digital Evidence

CHAIN OF CUSTODY

A process that tracks the movement of evidence through its collection, safeguarding, and analysis lifecycle by documenting each person who handled the evidence, the date/time it was collected or transferred, and the purpose for the transfer.

- COMPUTER SECURITY RESOURCE CENTER (CSRC)
National Institute of Standards and Technology (NIST)
https://csrc.nist.gov/glossary/term/chain-of-custody

CHAIN OF CUSTODY FORM

- Documented by a relevant form (***chain-of-custody form***)
- A ***documented record*** of ***who*** had ***possession and control over*** a particular piece of ***evidence*** at ***every moment until*** that object is entered into ***evidence in the courtroom***.
- ***Contributes to authenticity*** of these records

Property Record Number: _____

Anywhere Police Department
EVIDENCE CHAIN OF CUSTODY TRACKING FORM

Case Number: _____ Offense: _____
Submitting Officer: (Name/ID#) _____
Victim: _____
Suspect: _____
Date/Time Seized: _____ Location of Seizure: _____

Description of Evidence		
Item #	**Quantity**	**Description of Item** (Model, Serial #, Condition, Marks, Scratches)

Chain of Custody				
Item #	**Date/Time**	**Released by** (Signature & ID#)	**Received by** (Signature & ID#)	**Comments/Location**

APD_Form_#PE003_v.1 (12/2012) Page 1 of 2 pages (See back)

Chain-of-custody form (Source: "Sample Chain of Custody Form," NIST. Retrieved from http://www.nist.gov/oles/forensics/upload/Sample-Chain-of-Custody-Form.docx.)

EVIDENCE COLLECTION

Types of Evidence

Inculpatory evidence

- Evidence *show defendant is guilty* or had criminal intent
- *Supports our initial hypothesis* about what may have happened

Exculpatory evidence

- Statement *tending to excuse, justify, or absolve the alleged fault* or guilt of a defendant
- Disproves or *fails to support our current hypothesis*

EVIDENCE COLLECTION

Ways to acquire evidence

Live collection

- *Computer running* the operating system (OS)
- *Data collection goes on* using the resources of the computer itself
- Investigators logged in to machine

Dead collection

- Utilize *an OS that is not running*
- *Access the persistent storage* of the computer
- *Lost the volatile information*. E.g. memory, network connections, running processes, etc.

INVESTIGATION ROLES

First Responder

- Identifies and protects crime scene
- Preserves volatile evidence

Investigator

- Establish chain of command
- Conduct search of incident scene
- Maintain integrity of evidence

Crime Scene Technician

- Preserves volatile evidence and duplicates disks
- Shutdown systems for transport
- Tags and logs evidence
- Package and transport evidence
- Processes evidence

Adapted from Figure 3.1 Brooks, C. L. (September 2014). CHFI Computer Hacking Forensic Investigator Certification All-in-One Exam Guide. McGraw-Hill.

INVESTIGATION METHODOLOGY

Acquisition

- Gather information and evidence
- Make exact copy of computer data (volatile/non-volatile)

Authentication

- Ensure acquired evidence same as original data
- Inspect evidence to be used in court or investigation

Analysis

- Ensure duplicated data not modified
- Examine and evaluate information

DEFINE A FORENSIC PROCESS

1 Pre-Investigation	2 Preparing the Investigation	3 Search and Seizure	4 Analysing the evidence	5 Reporting and Testifying	6 Post-Investigation
Inputs					
Requirements for Forensic Capability	Information about the case	Plan for Obtain Evidence	Preserved Evidence	Analysis Results	Final Report, Evidence Notes
Process					
Build Forensic Capabilities	Develop Hypotheses and Plans	Forensicall y Sound search and Seizure	Analysis based on Hypotheses	Write Final Report	Process
Output					
Forensic Hardware and Methodology	Plan for Obtaining digital Evidence	Preserved and documente d Evidence	Analysis Results with Probative Value	Final Report and Testimony	Updated Risks and Investigative Methods

Adapted from Figure 3.1 from Brooks, C. L. (September 2014). *CHFI Computer Hacking Forensic Investigator Certification All-in-One Exam Guide.* McGraw-Hill.

FORENSIC PROCESS

5 WH

Who?

What?

When?

How?

Where?

Why?

PRE-INVESTIGATION

(TYPES OF COMPUTER FORENSICS TOOLS)

Types of Forensic Tools

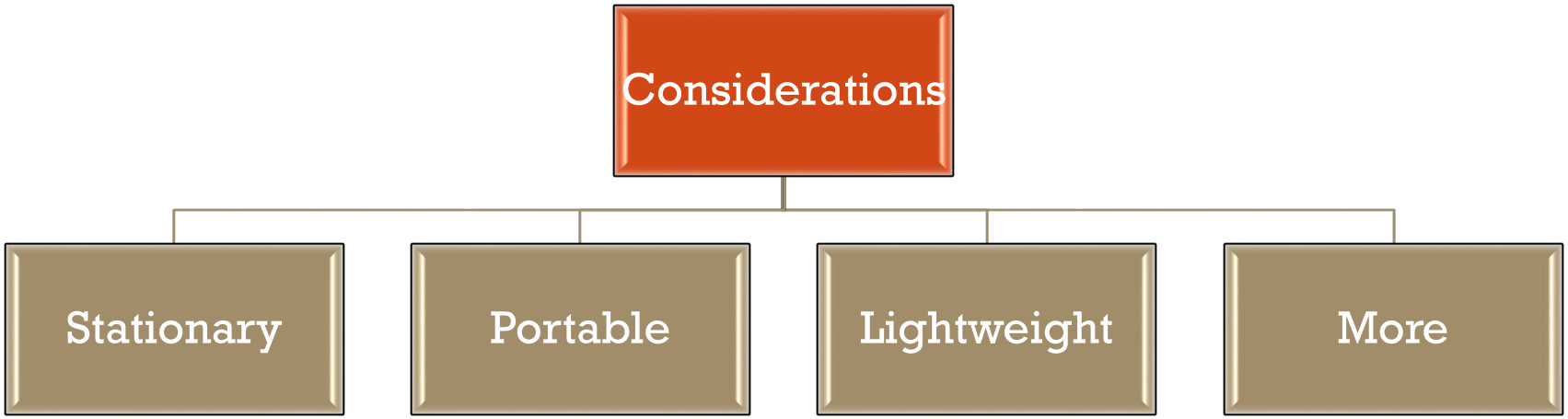
- **Software**
  - *Command-line/GUI applications*
  - E.g. AccessData FTK, Guidance Software EnCase, X-Ways Forensics
- **Hardware**
  - Range from *simple, single-purpose components* to complete computer systems and servers
  - Cables, connectors, Digital Intelligence F.R.E.D systems, DIBS Advanced Forensic Workstations

Supports?

- Acquisition
- Validation
- Discrimination
- Extraction
- Reconstruction
- Reporting

PRE-INVESTIGATION
(FORENSIC TOOLS CONSIDERATIONS)

Carefully consider what you need



What you need

What your system can handle

Balance?

What about ?

- Flexibility
- Reliability
- Expandability

PRE-INVESTIGATION

(TYPES OF COMPUTER FORENSICS SOFTWARE)

Advantages	Disadvantages
• Require few system resources • Run in minimal configurations • Fit on a bootable disk	• Cannot search inside archive and cabinet files • Most of them only work on file systems

Evidence could be admitted in court?

Test and validate your software to prevent damaging the evidence

PRE-INVESTIGATION

(TYPES OF COMPUTER FORENSICS SOFTWARE)

Forensics Software

- **Command -Line**
- **GUI**

- Simplify computer forensics investigations
- Help training beginning investigators
- Most of them come into suites of tools

- Mainly the *nix platforms
- E.g. The Coroner's Toolkit (TCT), The Sleuth Kit (TSK), Autopsy

E.g. EnCase, FTK, Norton DiskEdit, etc.

Advantages	Disadvantages
• Ease of use • Multitasking • No need for learning older OSs	• Excessive resource requirements • Inconsistent results • Tool dependencies

MORE

PRE-INVESTIGATION

(COMPUTER INVESTIGATION LAB)

Types of Labs

Police Agency Labs

- Need many options
- Use several PC configurations

Private Corporation Labs

- Handle only system types used in the organization

Keep a hardware library

PRE-INVESTIGATION

(INITIAL RESPONSE FIELD KIT)

Computer forensics kit

- Hardware
- Software
- Documentation Tools

Evidence Photography

- Document Crime Scenes

Figure 5-5 Items in an initial-response field kit

Forensics Workstation

- Forensics Workspace

Additional Lightsource for dark corners

MORE

Figure 5-5 Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

PRE-INVESTIGATION

(COMPUTER INVESTIGATION WORKSTATIONS)

Provide analysis capabilities

Build your Own Workstation

+	−
Customized to your needs	Hard to find support for problems
Save money	Can become expensive if careless
Broadband	

Buy From vendor

Examples:

- **F**orensic **R**ecovery of **E**vidence **D**evice (F.R.E.D)
- Tableau Forensics (FIRE IDE)

Hardware eventually fails

How?

Anticipate failures and schedule replacement

MORE

PRE-INVESTIGATION

(EXAMPLE OF FORENSIC ITEMS)

Write-Blocker

- Prevents data writes to a hard disk
- Discards the written data
- Hardware/Software

Faraday Isolation Bag

- Radio frequency–shielding material
- Prevent communication with devices

"Portable forensic tableau" by ErrantX from Wikimedia Commons is licensed under public domain
"Faraday bag" by Fullmetalwikial from Wikimedia Commons is licensed under Creative Commons Attribution-Share Alike 4.0 International

MORE

PREPARING THE INVESTIGATION

(INCIDENT/CRIME SCENES)

Types of Incident Scenes

Permission? Authorization?

Private-Sector

- **Authorized by senior management**
  - Reasonable suspicion
  - State when an investigation can be initiated
- **Employees' expectation of privacy**
  - Publish a privacy policy
  - Use warning banners

Law Enforcement

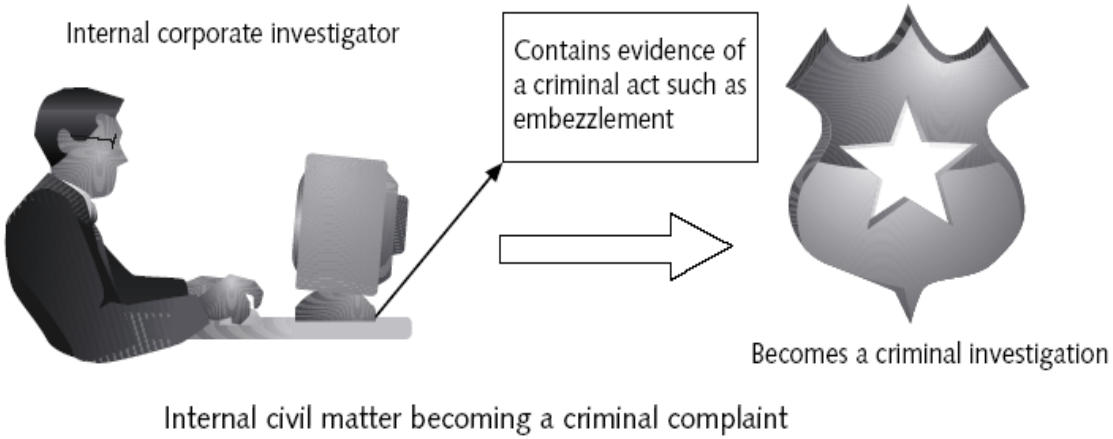
- **Criminal rules of search and seizure**
- **Probable cause**
  - Committed specific crime
  - Existence of evidence
  - Scope of search (Included Items)
- **Warrant**
  - Probable cause
  - Witness

PREPARING THE INVESTIGATION

(INCIDENT/CRIME SCENES)

Corporate environment is much easier than criminal environment

Private-sector investigation can become a criminal complaint



Check with corporate attorney

- Commingled data
- Warrants
- Subpoena
  - Witness summons
- Civil liability

Figure from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

PREPARING THE INVESTIGATION
(SAMPLE OF SEARCH WARRANT)

Seize and examine, by persons qualified to do so, and in a laboratory setting, any and all electronic data processing and computer storage devices, including; central processing units, internal and peripheral storage devices such as fixed disks, external hard disks, floppy disk drives and diskettes, tape drives and tapes, optical storage devices, optical readers and scanning devices, CD Rom drives and Compact Disks and related hardware, digital cameras and digital storage media, operating logs, software and operating instructions or operating manuals, computer materials, software and programs used to communicate with other terminals via telephone or other means, and any computer modems, monitors, printers, etc., that may have been used while engaging in [specify the illegal conduct], as defined in the Annotated Code of Maryland, amended and revised.

Extract Figure 5-4 Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

MORE

PREPARING THE INVESTIGATION
(SEARCH PREPARATION)

Identifying the nature of the case
Public or Private?

Identifying the type of computer system
Target Hardware/Software

Determining whether you can seize a computer
Warrant?

Obtaining a detailed description of the location
Potential Hazards?

Determining who is in charge
Law Enforcer/ Corporate Investigator

Using additional technical expertise
Need Specialist?

Determining the tools you need
Hardware/Software

Preparing the investigation team
Review plan with team

SEARCH AND SEIZURE

(ON THE SCENE)

Labeling Everything and Taking Pictures → **Identifying Sources** → **Seizing the Evidence**

- Physically label and document everything
- E.g. input ports, cables, devices, and others
- Capture image of computer screen

- Identify possible data source
- E.g. Computer, External disk drive, a USB stick, a network connection, etc.

- Preserve evidence
- Chain-of-evidence form
- Chain-of-custody form

How?

Environmental Impact on evidence

High or Low Humidity?

Static Discharge?

Condensation?

SEARCH AND SEIZURE
(SECURE THE SCENE)

Detach computers from Internet when necessary
Limit physical access and remote access

Volatile Data

- Data or other evidence lost once power is lost
- Preserve for immediate examination
- Document anything that is visible

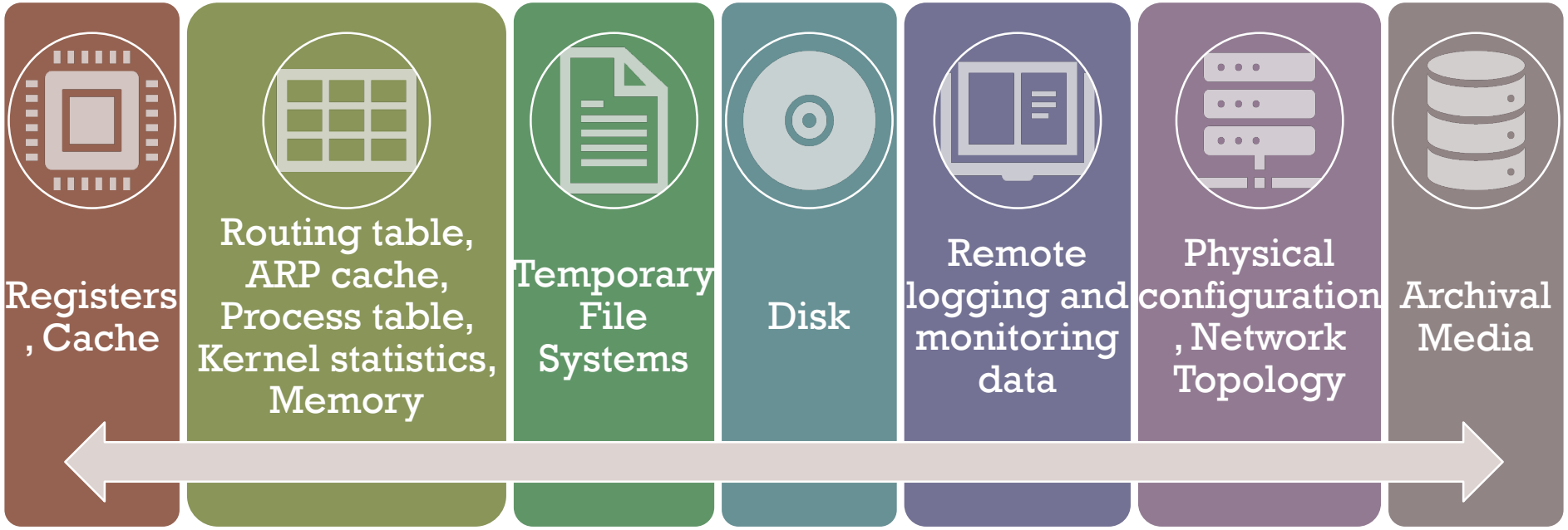
Non-Volatile Data

- If system is on, image computer's hard disk before power down
- If system is off, do not start it up
- Create bit-stream image of storage

Guidelines for Evidence Collection and Archiving
https://tools.ietf.org/html/rfc3227

SEARCH AND SEIZURE

(SECURE THE SCENE)



More Volatile → **Less Volatile**

Collect from more volatile to less volatile

Guidelines for Evidence Collection and Archiving

https://tools.ietf.org/html/rfc3227

SEARCH AND SEIZURE
(EVIDENCE CUSTODY LOG)

Item description:				
Item tag number:				
Person	Date logged out	Time logged out	Date logged in	Time logged in

Figure 5-7 A sample log file

Figure 5-7 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

SEARCH AND SEIZURE
(SECURE THE SCENE)

Maintain the integrity of digital evidence

Cyclic Redundancy Check (CRC) or Hashing (Digital Hash)

CRC

- Error-Detecting code
- Not considered a forensic hashing algorithm

Hashing

- Function to map data of arbitrary size to fixed-size
- Single bit changed, produce a different hash value
- E.g. MD5 , SHA-1, SHA-2 (Recommended), SHA-3

SEARCH AND SEIZURE
(SECURE THE SCENE)

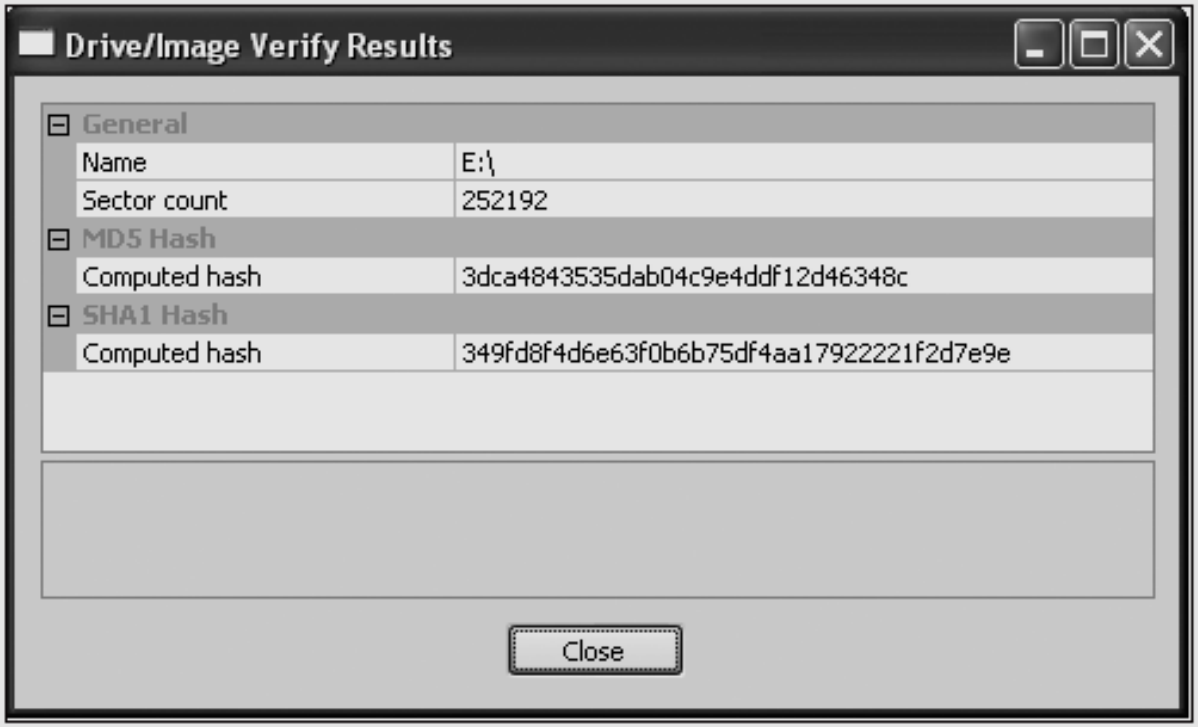


Figure 5-8 Using FTK Imager to verify hash values

Extract Figure 5-8 Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

DOCUMENTING EVIDENCE

Create an evidence custody form

- Identifies the evidence
- Identifies who has handled the evidence
- Lists dates and times the evidence was handled

Include any necessary information

- Add more information to your form might be needed
- E.g. Digital Hash, MD5, SHA-1, SHA-2 hash values, etc.

Evidence bags include labels and forms

- Evidence bags also include labels or evidence forms you can use to document your evidence

Figure 5-7 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

SEARCH AND SEIZURE *(THINGS TO AVOID)*

Don't shutdown until evidence collection is completed.
Attacker may alters startup/shutdown to destroy evidence

Don't trust programs on system.
Run programs from protected media

Don't run program that modify access files of systems

Guidelines for Evidence Collection and Archiving

https://tools.ietf.org/html/rfc3227

ANALYSING THE EVIDENCE
(MAKING WORKING COPY)

Create multiple working copies of original evidence

- Copy must be forensically sound
- Avoid working on original digital evidence
- Exact duplicate of the original
- Maintain the integrity with CRC and Hashing
- SHA-2 should be used

ANALYSING THE EVIDENCE *(PERFORMING ANALYSIS)*

Physical analysis

- **File Craving**
  - Reassemble computer files from data fragments without filesystem metadata

Logical Analysis

- **File System Information**
  - Filenames and sizes
  - File types, and other metadata
  - Slack space, unallocated space, more

ANALYSING THE EVIDENCE

(PERFORMING ANALYSIS)

Four ways to analyze the evidence

Time-frame analysis

- File activity at a particular time
- Examines various system log files
- Correlate file system activities with other activities

Application and file analysis

- Contents of files
- Maps files to applications
- Maps application activity to file actions

Data-hiding analysis

- Hidden Information within
- The file system
- Portion of disk inaccessible to file system access

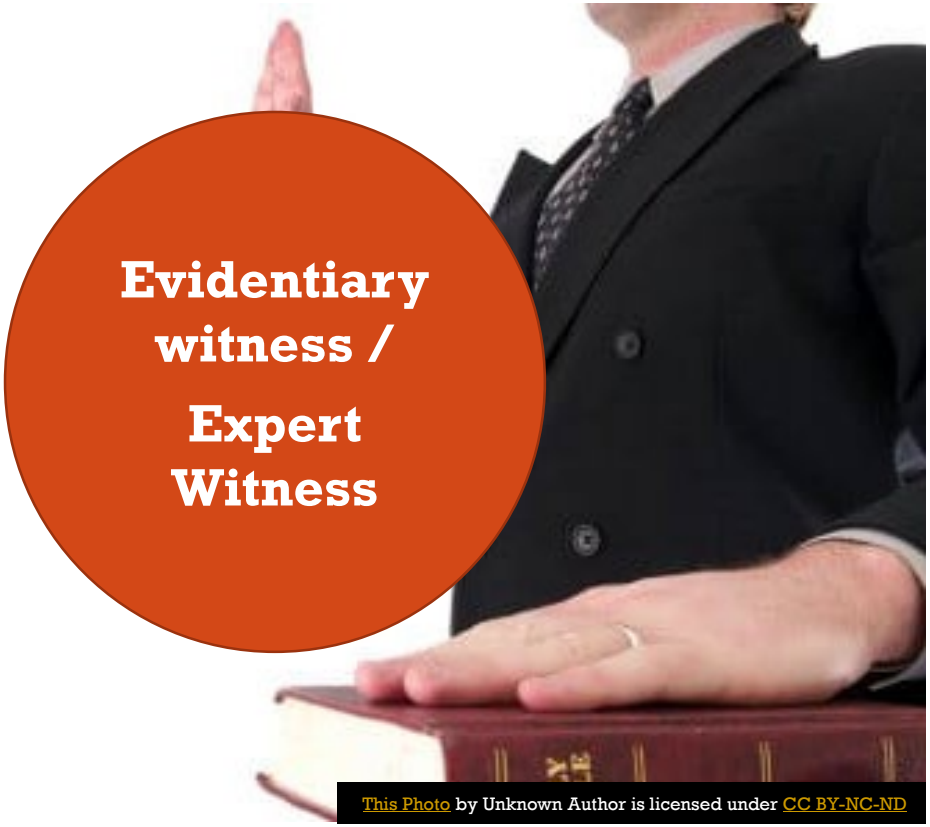
Ownership and possession analysis

- Identify activities tied to a particular login account

REPORTING AND TESTIFYING *(PREPARING THE FINAL REPORT)*

Findings	Final Report
• Systematic • Thorough • Unbiased	• Deliberate • Organized • Thoughtful

REPORTING AND TESTIFYING *(TESTIFYING IN COURT)*



This Photo by Unknown Author is licensed under CC BY-NC-ND

- Based on your background, experience, and qualifications,
- Testimony restricted to the facts
- Testify regarding your opinions about the evidence
- Requires special qualifications and job history

POST-INVESTIGATION *(CLOSING CASE)*

- Keep the case confidential
- Perform a postmortem analysis
- Replenish your forensics toolkit
- Archive all the evidence, reports, and other artifacts results
- Forensically sterilize any secondary media

CONCLUSION

- Digital Evidence
- Forensic Process
- Preparation for Investigation
- Search and Seizure during Investigation
- Evidence Analysis of Investigation
- Investigation Reporting, Testifying and Closure

REFERENCES

- Brooks, C. L. (September 2014). *CHFI Computer Hacking Forensic Investigator Certification All-in-One Exam Guide.* McGraw-Hill.
- Kleiman, D. (August 2011). The Official CHFI Study Guide (Exam 312-49). Syngress.
- Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.
- Carrier, B. (2005). *File System Forensic Analysis.* Addison-Wesley Professional.
- Messier, R. (2015). Operating System Forensics. Syngress.

EXTRA SLIDES

Table 5-1 Tools in an initial-response field kit

Number needed	Tools
1	Small computer toolkit
1	Large-capacity drive
1	IDE ribbon cable (ATA-33 or ATA-100)
1	SATA cable
1	Forensic boot media containing your preferred acquisition utility
1	Laptop IDE 40- to 44-pin adapter, other adapter cables
1	Laptop computer
1	FireWire or USB dual write-protect external bay
1	Flashlight
1	Digital or 35mm camera with film and flash
10	Evidence log forms
1	Notebook or dictation recorder
10	Computer evidence bags (antistatic bags)
20	Evidence labels, tape, and tags
1	Permanent ink marker
10	External USB devices, such as a thumb drive, or a larger portable hard drive

Table 5-1 Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

BACK

Table 5-2 Tools in an extensive-response field kit

Number needed	Tools
Varies	Assorted technical manuals, ranging from OS references to forensic analysis guides
1	Initial-response field kit
1	Portable PC with SCSI card for DLT tape drive or suspect's SCSI drive
2	Electrical power strips
1	Additional hand tools, including bolt cutters, pry bar, and hacksaw
1	Leather gloves and disposable latex gloves (assorted sizes)
1	Hand truck and luggage cart
10	Large garbage bags and large cardboard boxes with packaging tape
1	Rubber bands of assorted sizes
1	Magnifying glass
1	Ream of printer paper
1	Small brush for cleaning dust from suspect's interior CPU cabinet
10	USB thumb drives of varying sizes
2	External hard drives (200 GB or larger) with power cables
Assorted	Converter cables
5	Additional assorted hard drives for data acquisition

Table 5-2 Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

BACK

USING A WRITE-BLOCKER

- Prevents data writes to a hard disk
  - Discards the written data
  - For the OS, the data copy is successful
- Software options:
  - Software write-blockers are OS-dependent , e.g. PDBlock
- Hardware options
  - Ideal for GUI forensic tools
  - Act as a bridge between the disk and the workstation
  - Connecting technologies
  - FireWire, USB, SCSI controllers
  - e.g. www.voomtech.com, https://www.cru-inc.com/data-protection-topics/write-blockers/

BACK

RECOMMENDATIONS FOR A FORENSIC WORKSTATION

- Data acquisition e.g. USB, FireWire, Network
- Expansion devices requirements
- Power supply with battery backup
- External FireWire and USB ports
- Extra power and data cables
- High-end video card
- Ergonomic considerations e.g. Keyboard, mouse, Display
- Monitor

BACK

USING NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST) TOOLS

- Computer Forensics Tool Testing (CFTT) program
  - Based on standard testing methods
  - ISO 17025 criteria
  - ISO 5725
- Also evaluate disk imaging tools
  - Forensic Software Testing Support Tools (FS-TSTs)
- National Software Reference Library (NSRL) project – Note: From USA
  - Collects all known hash values for commercial software applications and OS files
  - Helps filtering known information

BACK

CONCEPTS AND TERMS IN WARRANTS

- **Innocent information**
  - Unrelated information that does not contribute to case
- **Limiting phrase**
  - Wording in search warrant that limits search scope of evidence
  - Separate innocent information from evidence
- **Plain view doctrine**
  - An exception to the Fourth Amendment's warrant requirement
  - Allows an officer to seize evidence and contraband that are found in plain view during a lawful observation
  - Searched area can be extended
- **Knock and announce**
  - Officers announce their authority and purpose before entering
  - Resident has reasonable opportunity to surrender premise

BACK

SAMPLE CIVIL INVESTIGATIONS

SAMPLE CIVIL INVESTIGATION

- Most cases in the corporate environment are considered noncriminal cases
- Common activities and practices
  - Recover specific evidence
    - Suspect's Outlook e-mail folder (PST file)
  - Covert surveillance
    - Its use must be well defined in the company policy
    - Risk of civil or criminal liability
  - **Sniffing** tools for data transmissions

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

SAMPLE CRIMINAL INVESTIGATION

- Computer crimes examples
  - Fraud
  - Check fraud
  - Homicides
- Need a warrant to start seizing evidence
  - Limit searching area

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

SAMPLE CRIMINAL INVESTIGATION (CONTINUED)

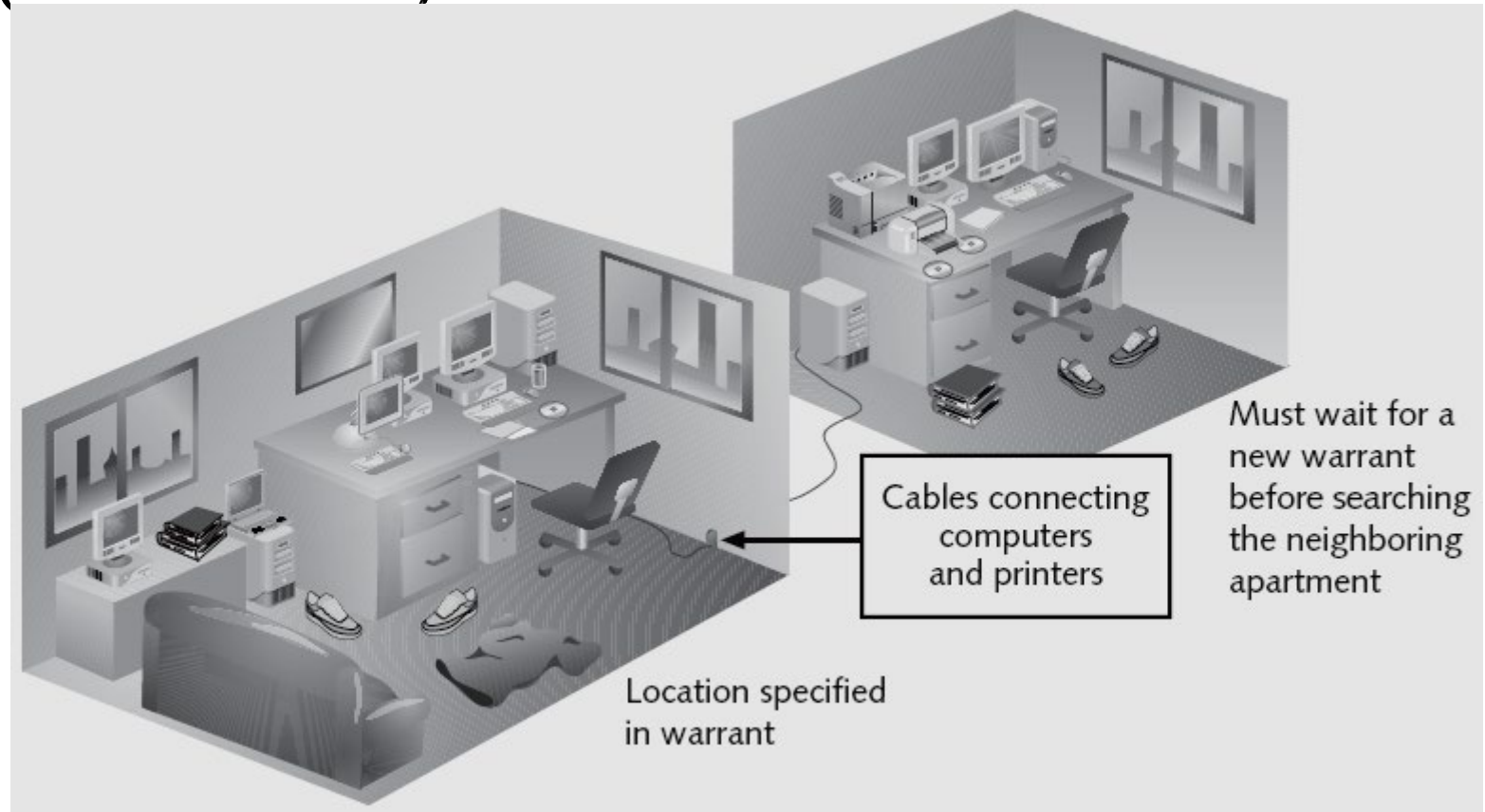


Figure 5-9 Search warrant limits

Figure 5-9 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

REVIEWING BACKGROUND INFORMATION FOR A CASE

- Company called Superior Bicycles
  - Specializes in creating new and inventive modes of human-driven transportation
- Two employees, Chris Murphy and Nau Tjeriko, have been missing for several days
- A USB thumb drive has been recovered from Chris's office with evidence that he had been conducting a side business using company computers

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

IDENTIFYING THE CASE REQUIREMENTS

Requirements

- Nature of the case
- Suspect's name
- Suspect's activity
- Suspect's hardware and software specifications

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

Conducting the Investigation: Acquiring Evidence with AccessData FTK

Assumptions/Knowledge

- Several incidents may or may not be related
- Suspect's computer can contain information about the case
- If someone else has used suspect's computer

Actions

- Extract the image from a bit-stream image file
- Analyze the image

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

CONDUCTING THE INVESTIGATION
(REFINE CASE IN FTK)

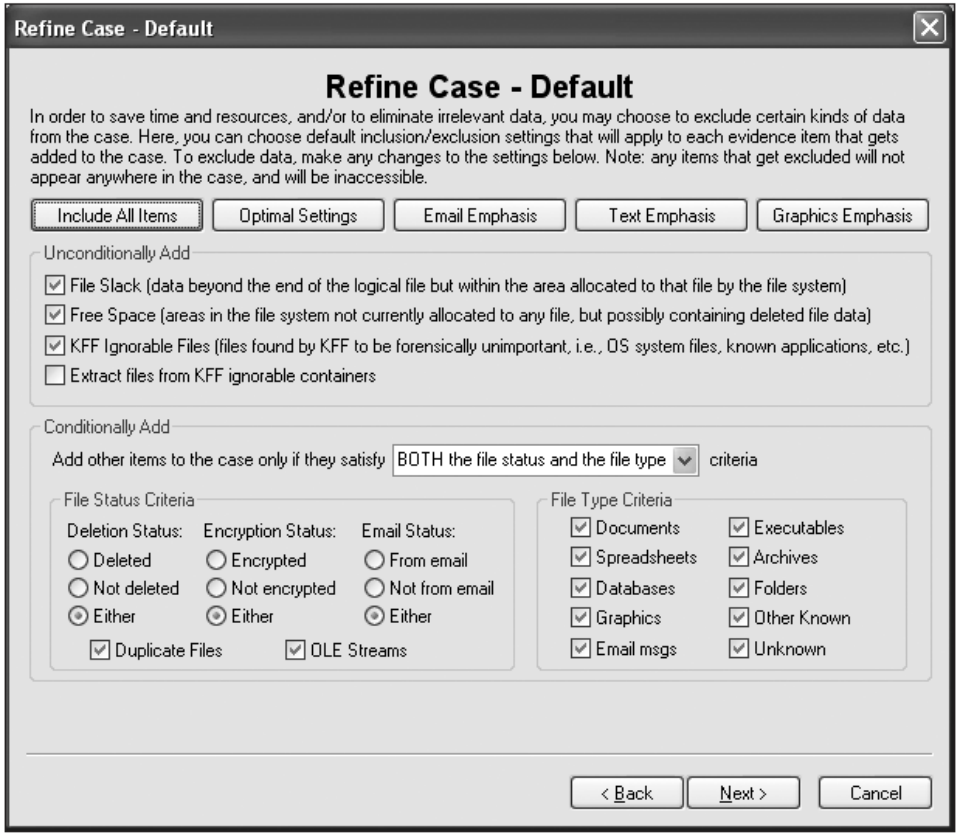


Figure 5-10 The Refine Case - Default dialog box

Figure 5-10 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

CONDUCTING THE INVESTIGATION
(EVIDENCE INFORMATION)

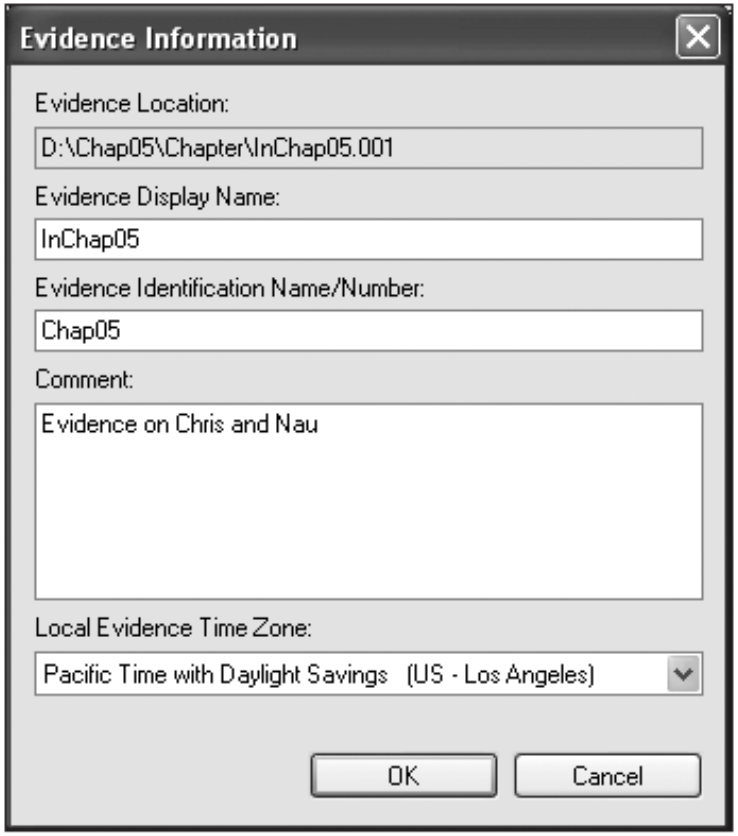


Figure 5-11 The Evidence Information dialog box

Figure 5-11 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

CONDUCTING THE INVESTIGATION
(ADD EVIDENCE)

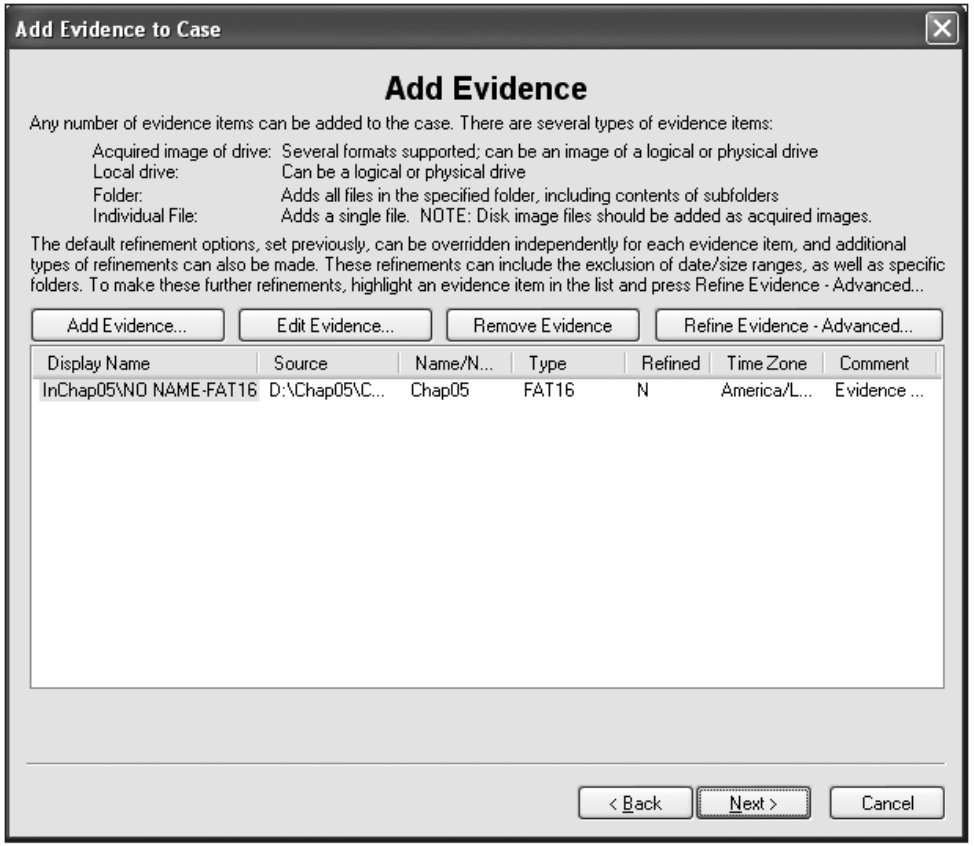


Figure 5-12 The Add Evidence to Case dialog box with image file listed

Figure 5-12 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

CONDUCTING THE INVESTIGATION

(SETUP NEW CASE)

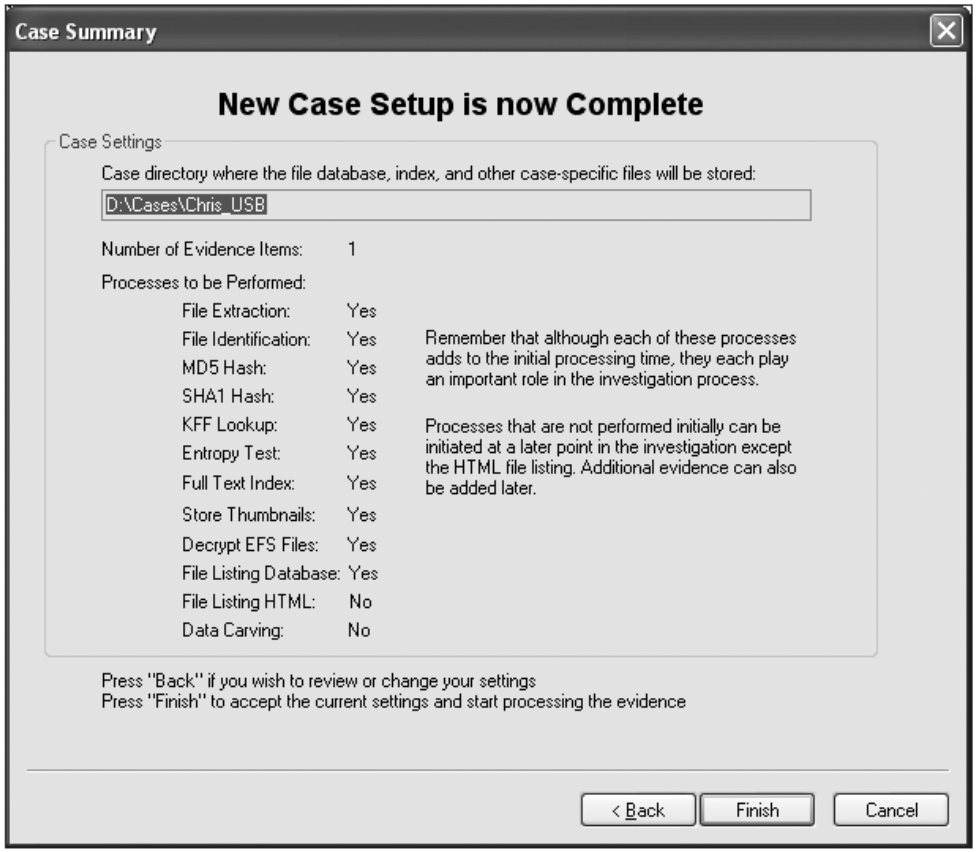


Figure 5-13 The Case Summary dialog box

Figure 5-13 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

CONDUCTING THE INVESTIGATION
(PROCESSING EVIDENCE)

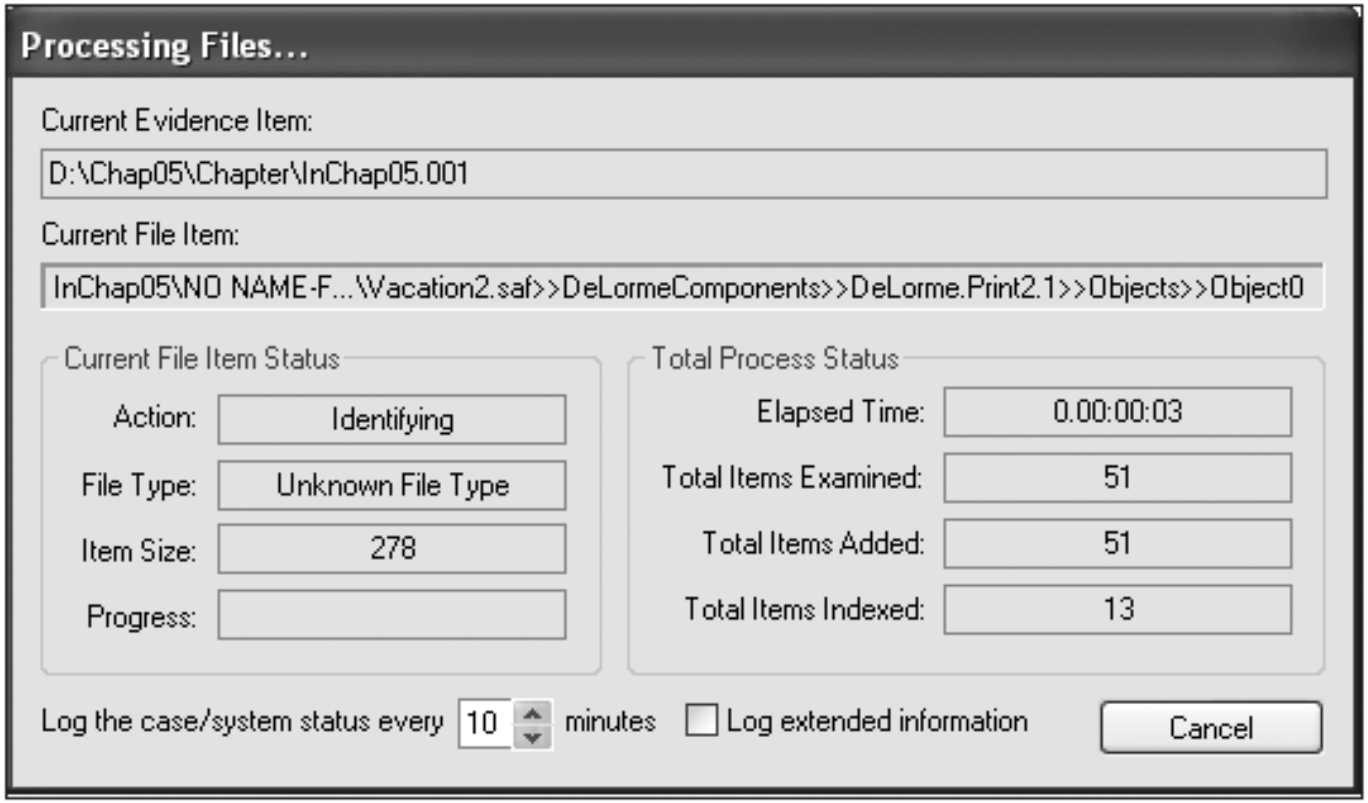


Figure 5-14 The Processing Files dialog box

Figure 5-14 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

CONDUCTING THE INVESTIGATION

(VIEW EVIDENCE OF INTEREST)

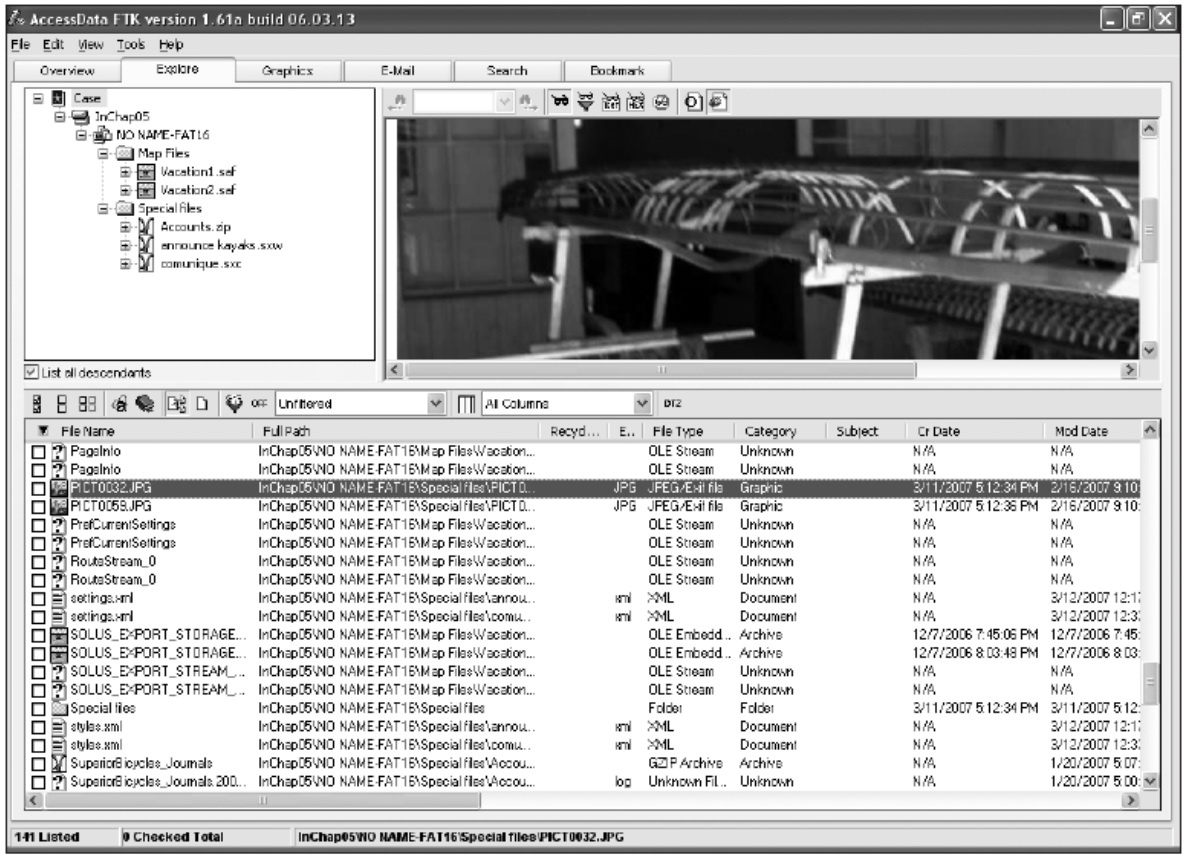


Figure 5-15 Selecting files of interest

Figure 5-15 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

CONDUCTING THE INVESTIGATION *(CREATE BOOKMARK)*

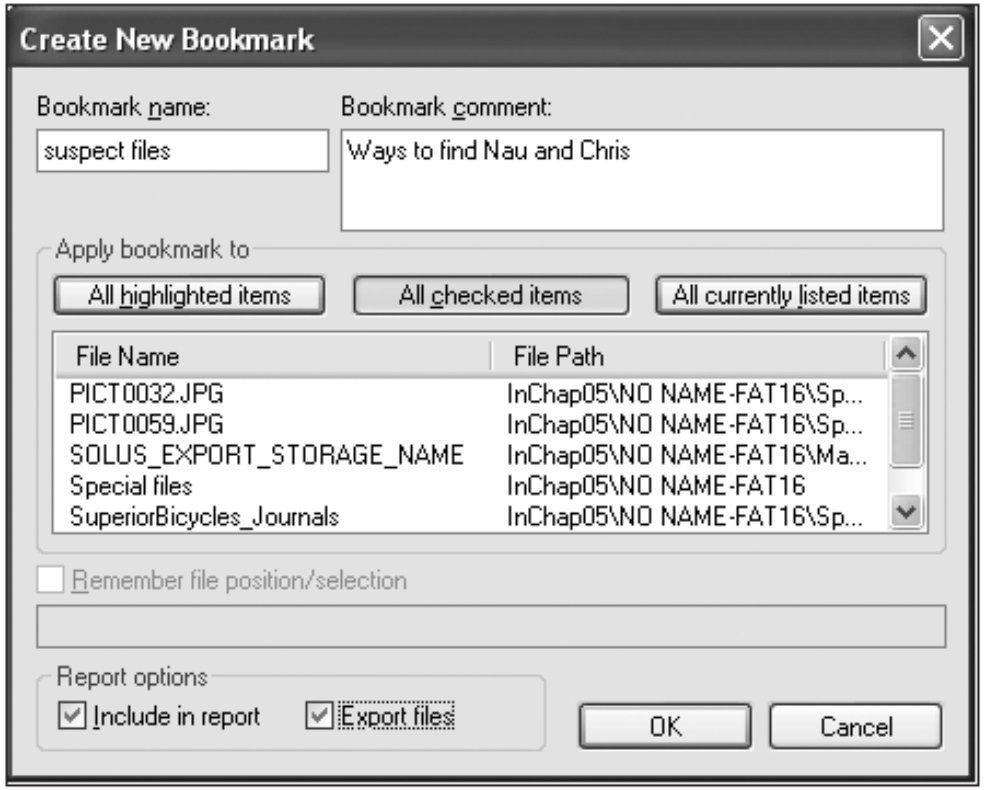


Figure 5-16 The Create New Bookmark dialog box

Figure 5-16 from Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.